

1. $\lim_{n \rightarrow \infty} \left(\sqrt[3]{n^3 + 2n^2} - \sqrt[3]{n^3 + 3n} \right) \left(\frac{3n^3 + 2n^2 - n}{n + 2n^3} \right)^3$

(A) $\lim \left(\sqrt[3]{n^3 + 2n^2} - \sqrt[3]{n^3 + 3n} \right) \cdot \lim \left(\frac{3n^3 + 2n^2 - n}{n + 2n^3} \right)^3$

(A) $\lim \left(n \cdot \sqrt[3]{1 + \frac{2}{n}} - n \sqrt[3]{1 + \frac{3}{n^2}} \right) \quad \times$

(A) $\lim \left(\sqrt[3]{n^3 + 2n^2} - \sqrt[3]{n^3 + 3n} \right) \cdot \frac{\overset{A^2}{\left(\sqrt[3]{n^3 + 2n^2} \right)^2} + \overset{AB}{\sqrt[3]{n^3 + 2n^2} \cdot \sqrt[3]{n^3 + 3n}} + \overset{B^2}{\left(\sqrt[3]{n^3 + 3n} \right)^2}}{\left(\sqrt[3]{n^3 + 2n^2} \right)^2 + \sqrt[3]{n^3 + 2n^2} \cdot \sqrt[3]{n^3 + 3n} + \left(\sqrt[3]{n^3 + 3n} \right)^2}$

$$\lim \frac{n^3 + 2n^2 - n^3 - 3n}{\sqrt[3]{n^3 + 2n^2}^2 + \sqrt[3]{n^3 + 2n^2} \cdot \sqrt[3]{n^3 + 3n} + \sqrt[3]{n^3 + 3n}^2} = \frac{2n^2 - 3n}{\underbrace{\quad}_{1}}$$

$$\lim \frac{n^2 \left(2 - \frac{3}{n} \right)}{n^2 \cdot \sqrt[3]{1 + \frac{2}{n}}^2 + n^2 \sqrt[3]{1 + \frac{2}{n}} \sqrt[3]{1 + \frac{3}{n^2}} + n^2 \sqrt[3]{1 + \frac{3}{n^2}}}$$

(A) $\lim \frac{2}{1+1+1} = \frac{2}{3}$

(B) $\lim \left(\frac{3n^3 + 2n^2 - n}{n + 2n^3} \right)^3 = \left(\lim \frac{3n^3 + 2n^2 - n}{n + 2n^3} \right)^3$

$$\lim \left(\frac{n^3 \left(3 + \frac{2}{n} - \frac{1}{n^3} \right)}{n^3 \left(2 + \frac{1}{n^2} \right)} \right)^3 = \left(\frac{3}{2} \right)^3 = \frac{27}{8}$$

$$(1) \quad A \cdot B = \frac{54}{24} = \frac{27}{12}$$

$$(2) \quad \lim_{n \rightarrow \infty} \sqrt[n]{\frac{6}{n^9}} \cdot \left(\frac{(n+1)^9 - (n-1)^9}{100 + 5n^3 + 3n^8} \right)^3$$

$$(A) \quad \lim_{n \rightarrow \infty} \sqrt[n]{\frac{6}{n^9}} \quad (B) \quad \lim_{n \rightarrow \infty} \left(\frac{(n+1)^9 - (n-1)^9}{100 + 5n^3 + 3n^8} \right)^3$$

$$(B) \quad \left(\lim_{n \rightarrow \infty} \frac{(n+1)^9 - (n-1)^9}{100 + 5n^3 + 3n^8} \right)^3 = \left(\lim_{n \rightarrow \infty} \frac{n^9 + \binom{9}{1}n^8 + \dots - (n^9 + \binom{9}{1}n^8 + \dots)}{n^8 \left(3 + \frac{5}{n^5} + \frac{100}{n^8} \right)} \right)^3$$

$$\left(\lim_{n \rightarrow \infty} \frac{n^8 \left(18 + \frac{1}{n} + \frac{1}{n^2} \dots \right)}{n^8 \left(3 + \frac{5}{n^5} + \frac{100}{n^8} \right)} \right)^3 = \left(\frac{18}{3} \right)^3$$

$$(A) \quad \lim_{n \rightarrow \infty} \sqrt[n]{\frac{6}{n^9}} = \lim_{n \rightarrow \infty} \frac{\sqrt[n]{6}}{\sqrt[n]{n^9}} = \lim_{n \rightarrow \infty} \frac{\sqrt[n]{6}}{n^{\frac{9}{n}} \cdot \sqrt[n]{1}} = 1$$

$$A \cdot B = \left(\frac{18}{3} \right)^3 = 216$$

$$\begin{aligned} & \uparrow \frac{1}{n} \ll 0 \\ & n^0 \cdot \sqrt[n]{1} \\ & 1 \cdot 1 = 1 \end{aligned}$$

$$(3.) \lim_{n \rightarrow \infty} \frac{n^{2n} - (n!)^2 + n^n \cdot \cos n}{3 \cdot n^n + 5 \cdot 100^{3n} - 2 \cdot n!}$$

$$\lim \frac{n^{2n} \left(1 - \frac{(n!)^2}{n^{2n}} + \frac{1}{n^n} \cdot \frac{\cos n}{n^{2n}} \right)}{3n^n \left(1 + \frac{5 \cdot 100^{3n}}{3 \cdot n^n} - \frac{2 \cdot n!}{3 \cdot n^n} \right)} = \frac{n^n \cdot \nearrow (1)}{3 \cdot \nearrow (1)}$$

$$\lim \frac{n^n}{3} = \infty$$

$$\begin{aligned}
 (4.) \quad \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n &= \lim_{n \rightarrow \infty} \left(\frac{n-1}{n}\right)^n = \lim_{n \rightarrow \infty} \frac{1}{\left(\frac{n}{n-1}\right)^n} \\
 &= \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n-1}\right)^{n-1} \left(1 + \frac{1}{n-1}\right)} = \frac{1}{e \cdot 1} = e^{-1}
 \end{aligned}$$

$\uparrow$
 e is definition

$\uparrow$
 0

$$5. \lim_{n \rightarrow \infty} \sqrt[n]{5^{2n+1} (2 + \cos n) + 6^n}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{5^{2n} \cdot \left(5 \cdot (2 + \cos n) + \frac{6^n}{5^{2n}} \right)}$$

$$\lim_{n \rightarrow \infty} 5^{\frac{2n}{n}} \cdot \sqrt[n]{5 \cdot (2 + \cos n) + \frac{6^n}{5^{2n}}}$$

$$\lim_{n \rightarrow \infty} 5^2 \cdot \sqrt[n]{10 + 5 \cdot \cos n + \frac{6^n}{5^{2n}}} \quad \leftarrow 0$$

↑
obmazan

$$\lim_{n \rightarrow \infty} 5^2 \cdot 1 = 25$$

$$6. \lim_{n \rightarrow \infty} \frac{\ln n - \frac{1}{2^n} + n! \cdot \sqrt{n}}{3n! + n^4 - \cos n} \left(\sqrt{n+5} - \sqrt{n+1} \right)$$

$$(A.) \lim_{n \rightarrow \infty} \frac{\ln n - \frac{1}{2^n} + \underline{n!} \cdot \sqrt{n}}{\underline{3n!} + n^4 - \cos n} \quad (B.) \left(\sqrt{n+5} - \sqrt{n+1} \right)$$

$$(B.) \left(\sqrt{n+5} - \sqrt{n+1} \right) \cdot \frac{\sqrt{n+5} + \sqrt{n+1}}{\sqrt{n+5} + \sqrt{n+1}}$$

$$\left(\frac{n+5 - n - 1}{\sqrt{n+5} + \sqrt{n+1}} \right) = \frac{4}{\sqrt{n} \cdot \left(\sqrt{1+\frac{5}{n}} + \sqrt{1+\frac{1}{n}} \right)}$$

$$= \frac{2 \cdot 2}{\sqrt{n} \cdot 2} = \frac{2}{\sqrt{n}} \quad \uparrow 0 \quad \uparrow 0$$

(A.)

$$\lim_{n \rightarrow \infty} \frac{\cancel{n!} \left(\frac{\ln n}{n!} - \frac{1}{2^n \cdot n!} + \sqrt{n} \right)}{\cancel{n!} \left(3 + \frac{n^4}{n!} - \frac{\cos n}{n!} \right)}$$

$0 \rightarrow \quad 0 \rightarrow \quad \infty$
 $0 \rightarrow \quad \text{increasing} / \infty = 0$

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n} + 0 + 0}{3 + 0 + 0} = \frac{\sqrt{n}}{3}$$

$$\lim_{n \rightarrow \infty} A \cdot B = \frac{\sqrt{n}}{3} \cdot \frac{2}{\sqrt{n}} = \frac{2\cancel{\sqrt{n}}}{3\cancel{\sqrt{n}}} = \frac{2}{3}$$

